

PROFILE

Innovation and R&D leader with 10+ years working on applied AI research (computer vision, robotics, autonomous perception, etc.) through to funded, industrially deployed programs. Experience leading multi-million euro (€4M+) R&D portfolios end-to-end: research strategy, opportunity identification, proposal development, multi-partner consortium delivery, and industrial rollout. Works at the intersection of technical research and funding strategy translating research priorities into fundable programs, coordinating stakeholders across research labs, industry partners, and technology teams, and representing programs externally. Bringing this background to EU funding strategy, proposal quality, and European partnership-building.

EXPERIENCE

Project Leader & Senior Research Engineer – AI Innovation Flanders Make

-  Apr 2022 – Present
-  Leuven, Belgium
- Lead large-scale AI R&D programs at national and EU level, including the 4M ALARMM SBO project involving 4 research groups and 7 industrial partners.
 - Represent the research group in national and European collaborations, contributing to strategic positioning and long-term research roadmaps.
 - Actively involved in business development activities: identification of new research and innovation opportunities, partner engagement, and co-definition of future project proposals.
 - Manage and coordinate cross-functional teams (3–12 members) across academia, industry, and public research stakeholders.
 - Drive development of advanced AI systems in computer vision, visual SLAM, and sensor fusion for industrial environments.
 - Achieved 4x technology transfer from research to industry in the most recent completed project (2025), enabling follow-up innovation initiatives.
 - Oversee full project lifecycle: opportunity scouting, ideation, consortium building, proposal writing, technical execution, stakeholder alignment, and technology transfer.
 - Mentor engineers and contribute to internal capability building, as well as strategic research and innovation roadmap definition.

Research Engineer – Computer Vision & AI Flanders Make

-  Jan 2018 – Mar 2022
-  Leuven, Belgium
- Designed AI-based vision systems for industrial inspection and automation.
 - Applied deep learning and self-supervised learning for real-time perception systems.
 - Contributed to EU-funded project proposals and technical work packages.

Research Engineer – Intelligent Vehicles Flanders’ DRIVE

-  May 2015 – Dec 2017
-  Lommel, Belgium

PhD Researcher – Embedded Vision for Robotics Institut Pascal

-  2011 – 2015
-  Clermont-Ferrand, France

CORE COMPETENCIES

AI & Perception Systems

- Computer Vision (2D/3D)
- Visual SLAM
- Sensor Fusion
- Deep Learning
- Pose Estimation
- Object Detection
- Quality Monitoring
- AR/VR for Industry

Tools Currently using & used before

- Python
- OpenCV
- Open3D
- ROS2
- Git
- Docker & Containerization
- CI/CD
- MLOPS
- PyTorch
- C/C++
- VHDL
- MATLAB
- JAVA

Emerging Focus - Learning

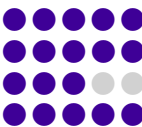
- AI Workflow Automation (n8n)
- AI Agents
- LLM Applications
- MCP

EDUCATION

- PhD in Electronics & System Architecture, Blaise Pascal University, (2011-2015), France
- MSc in Robotics & Multisensory Perception, Blaise Pascal University, (2010-2011), France
- Engineering Degree in Electronics, National Polytechnic School, (2005-2010), Algeria

LANGUAGES

- English
- French
- Dutch
- Arabic



PUBLICATIONS

 Full list of publications available on Google Scholar.

Journal Articles

- Weckx, Sam et al. (2022). "A cloud-based digital twin for monitoring of an adaptive clamping mechanism used for high performance composite machining". In: *Procedia Computer Science* 200, pp. 227–236.
- Birem, Merwan, Richard Kleihorst, and Norddin El-Ghouti (2017). "Visual odometry based on the Fourier transform using a monocular ground-facing camera". In: *Journal of Real-Time Image Processing*, pp. 1–10.
- Birem, Merwan and François Berry (2014). "DreamCam: A modular FPGA-based smart camera architecture". In: *Journal of Systems Architecture* 60.6, pp. 519–527.

Conference Proceedings

- Birem, M, C Domken, and A Bey-Temsamani (2024). "CAD2GraspMonitor: Vision-based Robotic Grasp Monitoring for Industrial Kitting Application". In: p. 129.
- Ruosch, C, M Birem, and A Bey-Temsamani (2024). "Bench Marking of Industrial Object Pose Estimation Algorithm Based on Vision & Deep Learning". In: p. 177.
- Solmaz, Serkan et al. (2024). "Capturing the system requirements for adopting Industry 5.0 in quality inspection: an evidence-based approach". In: vol. 128. Elsevier, pp. 369–374.
- André, Hugo et al. (2021). "Using a smartphone camera to analyse rotating and vibrating systems: Feedback on the SURVISHNO 2019 contest". In: vol. 154. Academic Press, p. 107553.
- Gors, Dorothy, Merwan Birem, et al. (2021). "An adaptable framework to provide AR-based work instructions and assembly state tracking using an ISA-95 ontology". In: vol. 104. Elsevier, pp. 714–719.
- Zogopoulos, Vasilios et al. (2021). "Image-based state tracking in augmented reality supported assembly operations". In: vol. 104. Elsevier, pp. 1113–1118.
- Dwek, Nathan et al. (2019). "Improving the accuracy and robustness of ultra-wideband localization through sensor fusion and outlier detection". In: vol. 5. 1. IEEE, pp. 32–39.
- Kikken, Edward et al. (2019). "A model-based approach to enhance performance in high-end laser-positioning systems". In: *2019 20th International Conference on Research and Education in Mechatronics (REM)*. IEEE, pp. 1–6.
- Gors, Dorothy, Robbert Hofman, et al. (2017). "New Object Tracker Based On Adaptive Intensity Models of Object and Its Surroundings". In.
- Shopovska, Ivana et al. (2017). "A hybrid fusion based frontal-lateral collaborative pedestrian detection and tracking". In: *2017 IEEE 20th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, pp. 1–6.
- Birem, Merwan, Jean-Charles Quinton, et al. (2014). "SAIL-MAP: Loop-closure detection using saliency-based features". In: *2014 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, pp. 4543–4548.
- Quétard, Boris et al. (2014). "Saliency in the fog: application to real and 2.5 D pictures". In: *TRA2014 Transport Research Arena 2014*.

PHILOSOPHY

"Life is like riding a bicycle. To keep your balance, you must keep moving."
– Albert Einstein

KEY PROJECTS

ALARMM SBO – Visual SLAM for Large Environments - [2025 – Present]

- Leading 4M multi-partner project on low-cost, accurate and robust vision SLAM systems
- Focus on robust perception in complex industrial environments
-  Latest project update: LinkedIn post - dissemination (April-2026)

QUALMA ICON – AI-Augmented Quality Assurance - [2022–2025]

- Led development of modular AI framework for large-scale manual assembly inspection
- Integrated computer vision (detection, pose estimation, tracking) with AR guidance
- Improved inspection accuracy and reduced cycle time in industrial scenarios
-  Demo video: QUALMA ICON Industrial YouTube Project Demonstration

FROGS ICON CAD2GraspMonitor – Robotic Grasp Validation - [2021–2023]

- Developed vision-based system for post-grasp validation in robotic kitting
- Combined YOLO-based detection, ICP pose estimation, and sensor fusion
- Ensured high reliability in industrial pick-and-place operations